RStudio

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multiple.r

```
1
2  data(mtcars)
3
4  plot(mtcars$mpg, type="l", col="blue",
5       xlab="Car Index", ylab="Values",
6       main="MPG and QSEC Line Chart")
7
8  lines(mtcars$qsec, col="red")
9
10 legend("topright",
11        legend=c("MPG", "QSEC"),
12        col=c("blue", "red"),
13        lty=1)
```

13:14 (Top Level) R Script

Console Terminal Background Jobs

R 4.5.1 · ~/

```
> source("~/DWDM/day2/multiple.r", echo = TRUE)

> data(mtcars)

> plot(mtcars$mpg, type="l", col="blue",
+      xlab="Car Index", ylab="Values",
+      main="MPG and QSEC Line Chart")

> lines(mtcars$qsec, col="red")

> legend("topright",
+        legend=c("MPG", "QSEC"),
+        col=c("blue", "red"),
+        lty=1)
>
```

Environment History Connections Tutorial

Import Dataset 141 MiB List

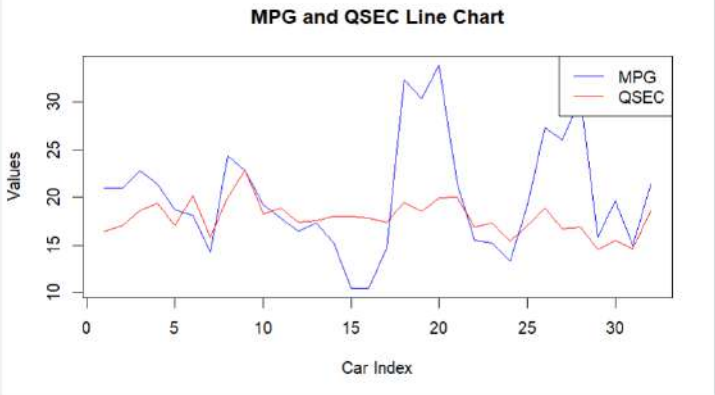
R Global Environment

Data

bin_boundaries	List of 4
bin_mean	List of 4
bin_median	List of 4
bins	List of 4
df	12 obs. of 2 variables
mtcars	32 obs. of 11 variables
slices	4 obs. of 2 variables
students	4 obs. of 2 variables

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RStudio

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```
1
2  install.packages("HSAUR")
3  library(HSAUR)
4
5  data("water", package="HSAUR")
6
7  plot(water$hardness, water$mortality,
8       main="Mortality vs Hardness",
9       xlab="Hardness", ylab="Mortality",
10      pch=19, col="blue")
11
12 model <- lm(mortality ~ hardness, data=water)
13
14 abline(model, col="red")
15
16 result <- predict(model, data.frame(hardness=88))
17 print(result)
```

5:31 (Top Level) R Script

Console Terminal Background Jobs

R 4.5.1 · ~/

```
> data("water", package="HSAUR")

> plot(water$hardness, water$mortality,
+      main="Mortality vs Hardness",
+      xlab="Hardness", ylab="Mortality",
+      pch=19, col="blue")

> model <- lm(mortality ~ hardness, data=water)

> abline(model, col="red")

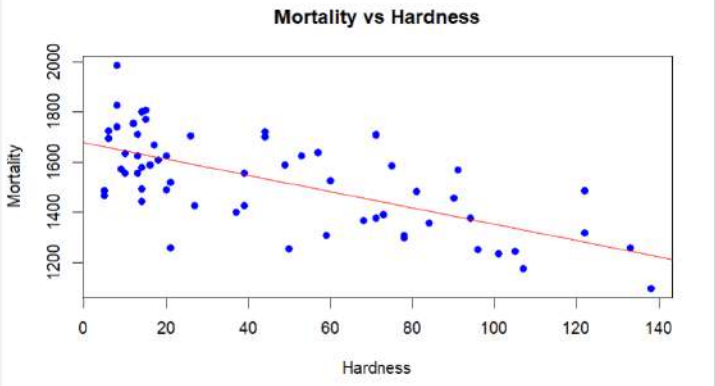
> result <- predict(model, data.frame(hardness=88))

> print(result)
     1
1392.46
```

Environment History Connections Tutorial

bin_mean	List of 4
bin_median	List of 4
bins	List of 4
df	12 obs. of 2 variables
model	List of 12
mtcars	32 obs. of 11 variables
slices	4 obs. of 2 variables
students	4 obs. of 2 variables
water	61 obs. of 4 variables

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```
1
2  data(mtcars)
3
4  boxplot(mpg ~ factor(cyl), data=mtcars,
5          main="MPG vs Number of Cylinders",
6          xlab="Number of Cylinders",
7          ylab="Miles Per Gallon",
8          col=c("lightblue", "lightgreen", "orange"))
```

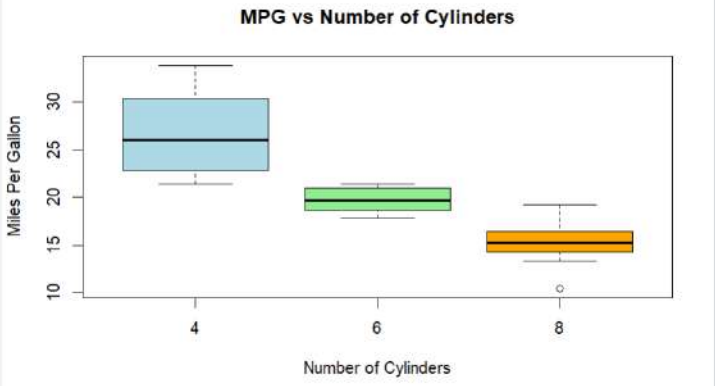
```
      1
1392.46
Warning message:
package 'HSAUR' was built under R version 4.5.3

> source("~/DWDM/day2/8.MFG.r", echo = TRUE)

> data(mtcars)

> boxplot(mpg ~ factor(cyl), data=mtcars,
+         main="MPG vs Number of Cylinders",
+         xlab="Number of Cylinders",
+         ylab="Miles Per ..." ... [TRUNCATED]
>
```

Environment	
bin_mean	List of 4
bin_median	List of 4
bins	List of 4
df	12 obs. of 2 variables
model	List of 12
mtcars	32 obs. of 11 variables
slices	4 obs. of 2 variables
students	4 obs. of 2 variables
water	61 obs. of 4 variables



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RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

```
1
2  data(mtcars)
3
4  boxplot(mpg ~ factor(cyl), data=mtcars,
5          main="MPG vs Number of Cylinders",
6          xlab="Number of Cylinders",
7          ylab="Miles Per Gallon",
8          col=c("lightblue", "lightgreen", "orange"))
```

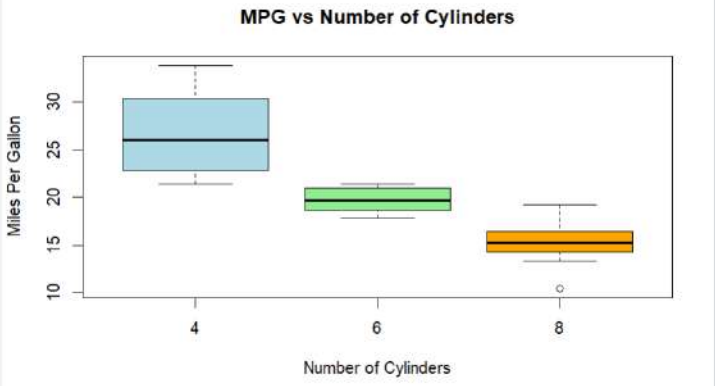
```
      1
1392.46
Warning message:
package 'HSAUR' was built under R version 4.5.3

> source("~/DWDM/day2/8.MFG.r", echo = TRUE)

> data(mtcars)

> boxplot(mpg ~ factor(cyl), data=mtcars,
+         main="MPG vs Number of Cylinders",
+         xlab="Number of Cylinders",
+         ylab="Miles Per ..." ... [TRUNCATED]
>
```

Environment	
bin_mean	List of 4
bin_median	List of 4
bins	List of 4
df	12 obs. of 2 variables
model	List of 12
mtcars	32 obs. of 11 variables
slices	4 obs. of 2 variables
students	4 obs. of 2 variables
water	61 obs. of 4 variables



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```
points <- c(10, 12, 13, 14, 15, 16, 17, 18, 20, 55)

boxplot(points,
        main="Tennis Players' Scores",
        ylab="Points Scored",
        col="lightblue")

print(boxplot.stats(points)$out)
```

```
+         main="MPG vs Number of Cylinders",
+         xlab="Number of Cylinders",
+         ylab="Miles Per ..." ... [TRUNCATED]
> source("~/DWDM/day2/9.outliners.r", echo = TRUE)

> points <- c(10, 12, 13, 14, 15, 16, 17, 18, 20, 55)

> boxplot(points,
+         main="Tennis Players' Scores",
+         ylab="Points Scored",
+         col="lightblue")

> print(boxplot.stats(points)$out)
[1] 55
>
```

median_value	9.5
min_age	23
min_mark	50
min_max	0.315789473684211
min_value	200
mode_value	"9"
modes	chr [1:2] "25" "35"
N	3194
pencils	num [1:10] 9 25 23 12 11 6 7 8 9 10
points	num [1:10] 10 12 13 14 15 16 17 18 20 55



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11qsfind.r | 3.classes.r | multiple.r | 7.linear.r | 8.MFG.r | 9.outliners.r | 10.diabetes.r | R data

```
1
2  data <- read.csv("~/DWDM/day2/diabetes.csv")
3
4  plot(data$Age, data$BloodPressure,
5       main="Blood Pressure vs Age",
6       xlab="Age", ylab="Blood Pressure",
7       col="blue", pch=19)
8
9  age_group <- cut(data$Age,
10                  breaks=c(0, 30, 40, 50, 100),
11                  labels=c("Below 30", "31-40",
12                           "41-50", "Above 50"))
13
14 avg_bp <- aggregate(BloodPressure ~ age_group,
15                     data=data, FUN=mean)
16
17 barplot(avg_bp$BloodPressure,
18         names.arg=avg_bp$age_group,
19         main="Average Blood Pressure by Age",
20         xlab="Age Group",
21         ylab="Average Blood Pressure",
22         col="lightgreen")
```

22:26 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.5.1 · ~/

```
+       col="blue", pch=19)

> age_group <- cut(data$Age,
+                  breaks=c(0, 30, 40, 50, 100),
+                  labels=c("Below 30", "31-40",
+                       .... [TRUNCATED]

> avg_bp <- aggregate(BloodPressure ~ age_group,
+                     data=data, FUN=mean)

> barplot(avg_bp$BloodPressure,
+         names.arg=avg_bp$age_group,
+         main="Average Blood Pressure by Age",
+         xlab="Age Group",
+     .... [TRUNCATED]
>
```

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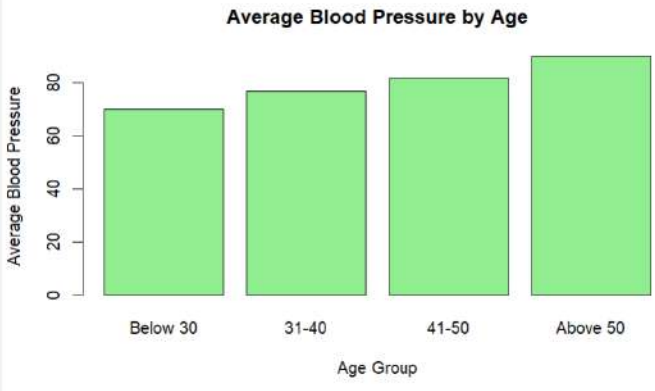
data	15 obs. of 2 variables
df	12 obs. of 2 variables
model	List of 12
mtcars	32 obs. of 11 variables
slices	4 obs. of 2 variables
students	4 obs. of 2 variables
water	61 obs. of 4 variables

Values

A num [1:9] 76 35 47 64 95 66 89 36 84

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